

UESgraphs: Automated graph-based district heating and cooling simulation model generation and analysis tool.

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Summary

UESgraphs (Urban Energy Systems graphs) is a Python-based tool for automated generation and analysis of graph-based district heating and cooling (DHC) network simulation models. The package enables researchers and engineers to create detailed thermal network models from minimal input data, including building characteristics, connection topologies, and thermal demands. A key strength of UESgraphs is its flexible data integration capability, supporting diverse input formats including OpenStreetMap data, GIS files (GEOJSON, Shapefiles), DWG files, CityGML datasets, as well as CSV files and Python dictionaries. Using graph theory principles, UESgraphs automatically constructs network representations where nodes represent buildings, substations, and supply stations, while edges represent pipes with associated thermal and hydraulic properties. A recent extension enables quasi-static simulations with pandapipes, providing a fast preliminary analysis option during the early planning phase before switching to detailed Modelica simulations for dynamic, network-level assessment. Key features include automated network topology simplification, hydronic sizing, automated quasi-static simulations using pandapipes (Lohmeier et al., 2020), automated Modelica model creation for thermo-hydraulic dynamic simulations, thermal loss calculations, and postprocessing of simulation results. Through its streamlined model development process, UESgraphs enables researchers and practitioners to perform rapid scenario analysis, system optimization, and informed decision-making in DHC network planning and operation.

Statement of need

DHC networks are essential infrastructure for achieving climate neutrality by decarbonizing the building sector (European Parliament and Council, 2023). However, planning and optimizing these networks requires complex thermo-hydraulic simulations that account for dynamic thermal losses, pressure drops, flow distributions, and temperature propagation through the network (Dénarié et al., 2023). Current simulation workflows present significant challenges that UESgraphs addresses through four key capabilities:

- **Automation** of DHC network model generation is critical for reducing development time and expertise barriers. Traditional approaches require manual creation of network topologies, pipe configurations, and component connections in simulation environments like Modelica/Dymola, consuming weeks of specialized effort and introducing errors (Wetter et al., 2014). UESgraphs automates the entire workflow from graph-based network topology to ready-to-simulate pandapipes (Lohmeier et al., 2020) and Modelica models, reducing development time from weeks to hours while ensuring consistency and reproducibility.

- 41 ■ **Integration** of graph theory with thermo-hydraulic simulation enables enhanced network
42 analysis and optimization. By representing DHC networks as graphs, nodes denote
43 components like buildings, substations, and supply stations, while edges represent pipes
44 with thermal and hydraulic properties. This structure enables UESgraphs to generate
45 simulation models and visualize simulation results directly within the graph framework.
46 Additionally the framework is used for automatic topology simplification, pipe sizing, and
47 network optimization. This graph-based approach facilitates rapid scenario comparison
48 and design alternative evaluation, essential for evidence-based DHC planning ([Schweiger
49 et al., 2017](#)).
- 50 ■ **Scalability** is a key requirement for resource-efficient transformation of the energy
51 sector, as numerous existing districts require retrofitting and thousands of new DHC
52 networks must be planned in the coming decades ([Connolly et al., 2014](#); [Persson et al.,
53 2014](#)). UESgraphs enables users to efficiently generate models for networks of varying
54 sizes and complexities, conduct comparative evaluations across multiple scenarios, and
55 apply findings to a wide range of district typologies. By providing a modular, open-
56 source framework, the tool supports the scientific community in addressing previously
57 underserved multidisciplinary challenges in urban energy modeling ([Reinhart & Cerezo
58 Davila, 2016](#)).
- 59 ■ **Accessibility** to advanced DHC simulation capabilities democratizes their use among
60 researchers, planners, and practitioners. The tool's Python-based architecture with
61 minimal input requirements (building characteristics, connection topology, thermal
62 demands) lowers the technical barrier for conducting detailed thermo-hydraulic analyses.
63 Automated post-processing functions enable quick evaluation of simulation results,
64 supporting iterative design processes and multi-criteria decision-making in DHC system
65 planning and optimization studies.

66 State of the field

67 Although several tools and frameworks already exist for modelling DHC systems, they typically
68 either focus on hydraulic steady-state analysis or necessitate significant manual input for dynamic
69 multi-physics simulations ([Kuntuarova et al., 2024](#); [Lund et al., 2014](#)). Modelica-based libraries
70 such as the Buildings library and AixLib provide detailed component models for pipes, heat
71 exchangers, and generation units, but they expect users to build the network topology explicitly
72 in the modeling environment ([Müller et al., 2016](#); [Wetter et al., 2014](#)). Outside the Modelica
73 ecosystem, tools such as pandapipes ([Lohmeier et al., 2020](#)) and related Python packages offer
74 hydraulic network analysis, but they typically do not couple with detailed thermo-hydraulic
75 pipe models for time-resolved simulations. The district model also has to be built manually
76 in these python based tools. Although commercial planning tools for district heating and
77 cooling offer powerful functionalities, they are often closed-source and difficult to integrate into
78 research workflows requiring scripted, reproducible model generation and analysis ([Kuntuarova
79 et al., 2024](#)). UESgraphs offers a unique set of capabilities in this field. The open-source
80 Python package integrates geospatial data and tabular data sources, constructs graph-based
81 representations, performs hydronic sizing and topology simplification, conducts quasi-static
82 pandapipes simulations, exports Modelica models and provides dedicated post-processing tools.

83 Software design: Functional principle

84 UESgraphs operates through a modular workflow that transforms minimal network topology
85 data into executable thermo-hydraulic simulation models. The tool follows a five-stage process
86 (Figure 1) : network definition, graph construction, topology simplification, model generation,
87 and post-processing analysis. The current implementation builds upon and extends previous
88 work on automated DHC network model generation ([Fuchs & Müller, 2017](#); [Mans et al., 2022](#)),

89 incorporating enhanced analysis capabilities, quasi-static simulation with pandapipes, expanded
90 input data interfaces, and includes post-processing features.

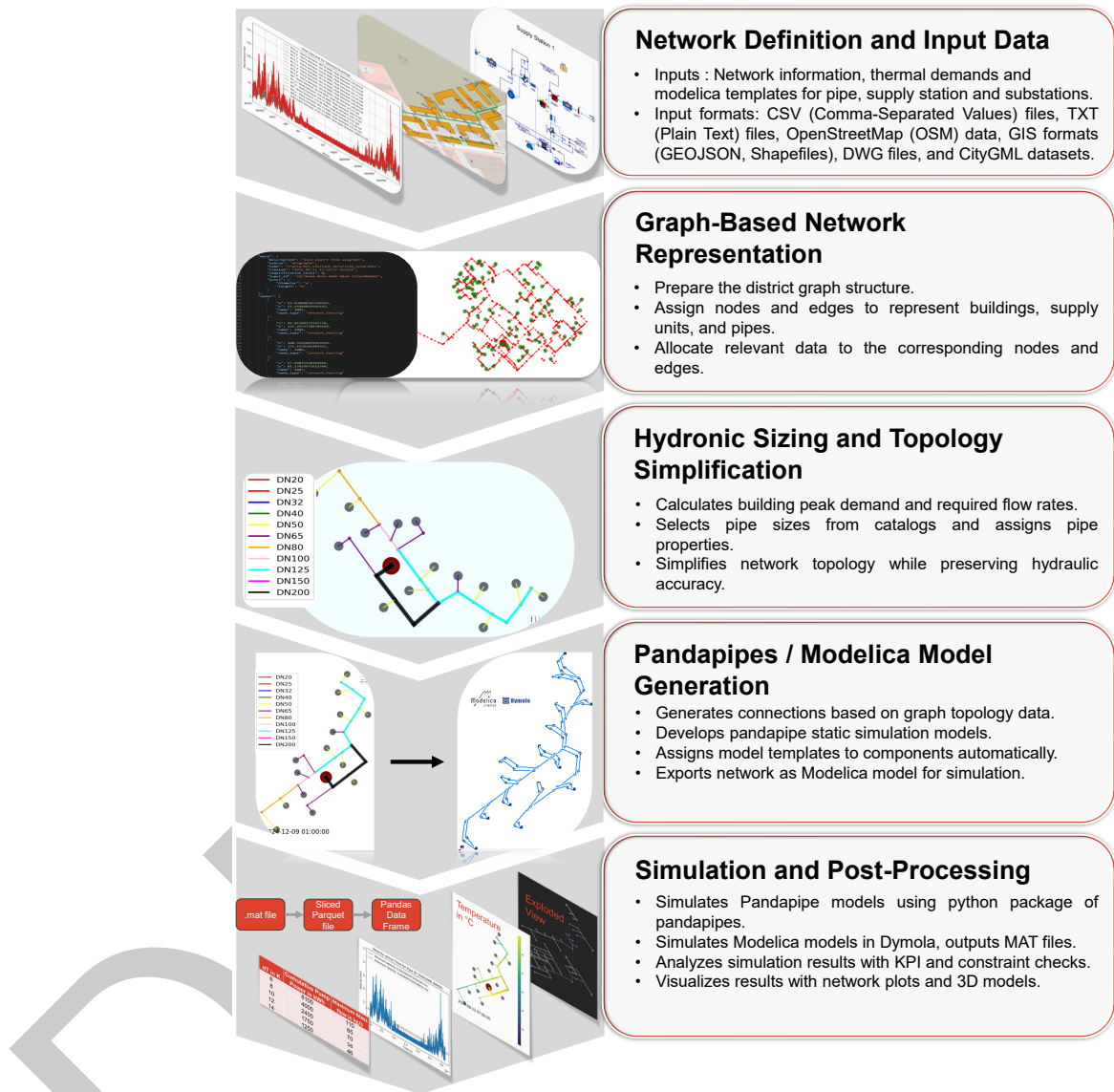


Figure 1: Workflow of UESgraphs showing the five-stage process from network definition to post-processing analysis.

91 **Network Definition and Input Data:** Users provide basic network information through CSV
92 files or Python dictionaries, specifying building locations, connection topology (graph edges),
93 thermal demand profiles, and optional supply/return temperature requirements. The minimal
94 input includes node coordinates, edge connections, and time-resolved heating or cooling
95 demands for each building. Network components such as substations, generation units, and
96 storage systems can be integrated through predefined templates. Rather than relying on a
97 single data source, UESgraphs provides input interfaces for OpenStreetMap data, GIS formats
98 (GEOJSON, Shapefiles), DWG files, and CityGML datasets.

99 **Graph-Based Network Representation:** The tool constructs a district graph using the NetworkX
100 library (Hagberg et al., 2008), where nodes represent buildings, heat sources, or network
101 junctions, and edges represent pipe segments. Each edge stores thermal-hydraulic properties

including pipe diameter, length, insulation thickness, and burial depth. The graph structure enables algorithmic network analysis, automatic detection of supply and return paths, and identification of critical network sections.

Automated Hydronic Sizing and Topology Simplification : UESgraphs extracts peak thermal demands from time-series data for each building given as input by the user, combining space heating and domestic hot water loads. The tool calculates required mass flow rates based on supply and return temperature differentials and determines corresponding flow velocities. Pipe diameters in UESgraphs are automatically selected from an integrated Isoplus manufacturer catalog, matching calculated flow requirements to standard pipe sizes so that flow velocities remain within user-specified thresholds (ISOPLUS Group, 2024). Network topology can be simplified through algorithmic merging of series-connected pipes or removal of redundant junction nodes while preserving hydraulic equivalence.

Quasi-Static Pandapipes Simulation and Modelica Model Generation: A dedicated module for converting the graph topology into a pandapipes (Lohmeier et al., 2020) model has now been included in the framework. This enables rapid static hydraulic analysis. This allows users to do preliminary checks on pressure drops, flow velocities and pump power before they commit to computationally expensive modelica based dynamic simulations. The core functionality exports the optimized network as ready-to-simulate Modelica models compatible with the open-source AixLib library (Müller et al., 2016). Users can either create custom Modelica models for supply systems, substations, and pipes, or utilize default models from the AixLib library package. The tool converts these Modelica models into templates, which are then systematically assigned to buildings, pipes, and supply components respectively. UESgraphs generates a complete district-level Modelica model by instantiating these templates with parameters automatically derived from the graph structure stored in JSON format.

Simulation and Post-Processing: The generated Modelica models are simulated in Dymola, producing result files in MAT format that contain time-resolved data for all network components. UESgraphs provides post-processing functions by handling the result files both from pandapipes and Modelica to analyze simulation outcomes, including constraint verification, Key Performance Indicator (KPI) evaluation, temperature distribution analysis, pressure drop assessment, thermal loss quantification, and energy balance verification across the network.

Research impact statement

UESgraphs is designed to make detailed DHC network modeling feasible in studies that require many scenarios or large networks by automating tasks that would otherwise have to be repeated manually in Modelica environments (Mans et al., 2022). By combining automatic hydronic sizing, topology simplification, quasi-static pandapipes simulations (Lohmeier et al., 2020) and template-based Modelica export, the tool allows researchers to construct and modify district-scale models in hours rather than weeks. This reduction in overhead directly supports parametric studies on temperature levels, insulation standards, pumping concepts, and control strategies, which are central questions in the transition toward fourth- and fifth-generation DHC systems (Lund et al., 2014; Mans et al., 2022). Integrating pandapipes strengthens the workflow by enabling quick quasi-static network simulations for early-stage design screening. This allows researchers to address specific planning questions before investing in detailed dynamic Modelica modeling. In the domain of applied research, UESgraphs has been adopted in the TransUrban.NRW living lab project, where it facilitates the planning and simulation of district heating and cooling concepts under real-world conditions (TransUrban.NRW Initiative, 2025). The combination of open-source availability on GitHub, continuous integration, and steadily improving documentation positions the tool as a community-ready platform for further methodological work on DHC modeling and optimization.

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AI usage disclosure

The authors developed the UESgraphs source code, including the core algorithms for graph handling, hydronic sizing, pandapipes simulation export and Modelica export, without relying on generative AI tools. However, generative AI tools (ChatGPT and Claude) were used to refine the wording, structure, grammar and clarity of the tool's documentation. Additionally, they were occasionally used to suggest code optimizations and generate repetitive boilerplate code segments, which were subsequently reviewed, adapted, and tested by the authors. All AI-assisted text and code were critically checked for technical accuracy and consistency with the actual implementation. The authors take full responsibility for both the code and the manuscript.

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